DS	**Session**	Linked List	**Question Information**	Level 1 Challenge 37

Question description

the popular engineering college got lowest pass percentage in last semester. the principal conducted faculty meeting and decided to visit all the classes surprisingly.

Dr.Ramprasath is a faculty, who handling data structure course for EEE department second year students.

one day this faculty was handling very interesting topic in data structure such that Linked List,

During this lecture time, principal surprisingly visited to the class and asking to conduct surprise test on Linked list concept.

So the faculty decided to conduct test on the topic of Linked List.

the question was given to last bench students that is,

The nodes are deleted before a certain given node in the linked list.

For example if the given Linked List is 5->10->15->20->25 and

delete before 15 then the Linked List becomes 15->20->25.

Constraint :

1< N < 1000

1< P < N-1

INPUT Format

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Third line contains position of the node to be deleted.

OUTPUT Format

Single line represents the final linked list after deletion.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4
9 77 12 6
12
```

EXPECTED OUTPUT

```
Linked List:->12->6
```

Test Case 2

INPUT (STDIN)

```
4
91 77 12 61
2
```

EXPECTED OUTPUT

```
Invalid Node! Linked List:->91->77->12->61
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
struct node
```

Test Case 2

KEYWORD

```
node *next;
```

Test Case 3

KEYWORD

```
void create()
```

Test Case 4	Test Case 5	Test Case 6
KEYWORD	KEYWORD	KEYWORD
`for(i=0;i<n;i++)`	`p1=p1->next`	`void del()`

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
3	270	58

Code

solution.c

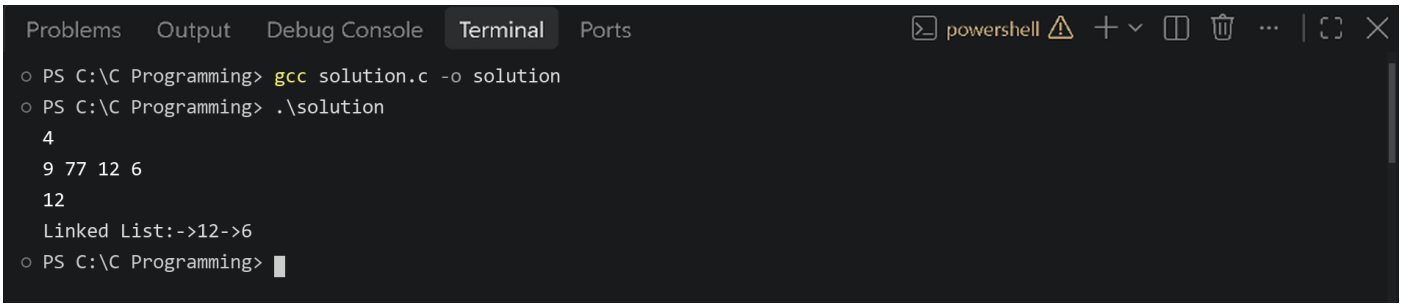
```c
#include <stdio.h>
#include <stdlib.h>

struct node {
    int data;
    struct node *next;
};

void create() {}
void del() {}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    struct node *head = NULL, *tail = NULL;
    for (int i = 0; i < n; i++) {
        int x; scanf("%d", &x);
        struct node *t = (struct node*)malloc(sizeof(struct node));
        t->data = x; t->next = NULL;
        if (!head) head = tail = t;
        else { tail->next = t; tail = t; }
    }
    int val;
    scanf("%d", &val);
    struct node *cur = head;
    while (cur && cur->data != val) {
        cur = cur->next;
    }
    if (!cur) {
        printf("Invalid Node! Linked List:");
        cur = head;
        while (cur) { printf("->%d", cur->data); cur = cur->next; }
        printf("\n");
    } else {
        printf("Linked List:->%d", cur->data);
        cur = cur->next;
        while (cur) { printf("->%d", cur->data); cur = cur->next; }
        printf("\n");
    }
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
4
9 77 12 6
12
Linked List:->12->6
PS C:\C Programming>